Seungmin Yome

M.S. Student in Energy & Petroleum Engineering

syome@uwyo.edu Laramie, WY, USA seungminyome

Research Experience

University of Wyoming, Graduate Research Assistant — Laramie, WY, USA — 2025 – present

- Built an axisymmetric finite-volume model (OpenFOAM + solids4Foam) of a spherical proppant grain loaded into and unloaded from Mohr–Coulomb rock through penalty contact; unloading separates permanent embedment from elastic recovery. All rock inputs are published Montney siltstone measurements; nothing is fitted to indentation data
- Compared the model with published Brinell indentation (35 N) and with eight proppant-pack embedment tests predicted blind; the pack comparison shows a grain-size-dependent mismatch that neither mesh refinement nor dilation angle removes, reported as a model limit rather than tuned away
- Showed that residual depth converges slowly with mesh and traced its mesh sensitivity to the gap between friction and dilation angles; every condition now runs on a single fine mesh, with its remaining mesh error carried as an uncertainty (≈2,000 runs on UW's MedicineBow cluster, Oct. 2026)
- Fitting dimensionless equations for both peak depth under load and residual depth after unloading, by leave-family-out cross-validation on a 551-condition design that balances the levels of every input, and mapping where each holds; froze predictions for new check conditions before seeing their results
- Pre-registered the test matrix and pass/fail checks (convergence, force balance) in Git before running, and automated the post-processing so every figure and statistic regenerates from the archived runs
- Trained in triaxial cell operation and DCI rock-compressibility (bulk modulus) testing
- Planned next: inverse models with prediction intervals to find which indentation measurement identifies the dilation angle; preparing the indentation experiments that will supply the data
- Designed and built a displacement-controlled micro-indentation rig for P–h curves on rock and metal (parametric CAD, 3D-printed fixtures, 1 mm tungsten-carbide ball) with a piezoelectric load cell and a laser Doppler vibrometer recorded through an NI DAQ in LabVIEW

KAIST — Graduate School of Mechanical Engineering, Research Intern (6-month full-time) — South Korea — 2024 – 2025

- Advisor: Dr. Yeunwoo Cho
- Sole investigator on a government-funded study of air-jet-driven waves on a thin elastic film: derived the dispersion relation with film stiffness and tension and confirmed the Kelvin-wave transition on high-speed video
- Assembled a free-surface cavitation tank from parts and explained the upper-wall contraction of ventilated supercavities with a wake-theory term in a 2D model's closure condition, closing most of the gap to the photos
- Simulated von Kármán vortex shedding near a free surface (ANSYS Fluent, VOF): the closer the surface, the lower the Strouhal number and the weaker the upper vortices

Pusan National University — Department of Naval Architecture & Ocean Engineering, Undergraduate Research Assistant — South Korea, 2022 – 2024

- Validated RANS CFD of KCS hull resistance against the Gothenburg 2010 benchmark (STAR-CCM+): 1.2% error in total resistance; responsible for mesh refinement at the bow, stern and free surface and for post-processing
- PIV and hull-resistance tests, ITTC 1957/1978 propulsion design, and an ANSYS Fluent study of an airfoil appendage on hull attitude and resistance
- Started the vibration/acoustic anomaly-detection model (MATLAB, Python) finished at Samsung

Industry Experience

Samsung Heavy Industries Laboratory, Mechanical Engineer Intern — South Korea, 2024 – 2024

- Trained an LSTM autoencoder on healthy-condition vibration and acoustic data only (Green Nuri vessel program) so anomalies appear as reconstruction error; detected 621 of 622 test anomalies, one traced to a weld fatigue crack

Pusan National University, Capstone Design Project — South Korea, 2023 – 2023

- Benchmarked engineering design tools for the shipbuilding industry (AVEVA Marine, TTM, VATIA, AutoCAD) across performance, integration, and usability criteria
- Proposed a cloud-based ML-integrated design platform (Google Cloud) balancing system scalability with market viability

Publications

Development of a dimensionless predictive equation for residual proppant embedment from finite-volume simulations of Montney siltstone — Jan 2026
Seungmin Yome, Soheil Saraji
(in preparation)

Skills

Simulation & CAE: solids4Foam, OpenFOAM, ANSYS Fluent, ABAQUS, STAR-CCM+, ParaView

Programming: Python, MATLAB, C++, LaTeX

Design & Engineering Tools: AutoCAD, AVEVA Marine, TTM, VATIA, Google Cloud Platform

Laboratory: Micro-indentation (rig built in-house), load cell, laser vibrometry, NI DAQ/LabVIEW, PIV, hull resistance testing; triaxial and DCI testing (trained)

Education

University of Wyoming, Energy & Petroleum Engineering — Laramie, WY, USA, 2025 – present

- Advisor: Dr. Soheil Saraji (Department Head)
- Minor: Computer Science (in progress)
- Thesis: Numerical modeling of proppant-supported fracture systems — embedment, rock deformation, and single-phase fluid transport using solids4Foam
- Coursework: Rock Mechanics, Enhanced Oil Recovery, Geostatistics, Python for Engineers
- Current coursework (Fall 2026): ML for Computational Solid Mechanics, Practical Machine Learning

Pusan National University, Naval Architecture & Ocean Engineering — South Korea, 2018 – 2024

- Advisor: Dr. Inwon Lee
- Coursework: Solid Mechanics, Fluid Mechanics, Linear Algebra, Thermodynamics, Ocean Wave Mechanics, FEM, Offshore Structural Design, CFD

Grants & Fellowships

Graduate Research Assistant, University of Wyoming (fully funded) 2025–Present
- University of Wyoming

Research Scholarship Jan 2024
- Korea Advanced Institute of Science and Technology

Undergraduate Research Student Grants 2022–2024
- Pusan National University

Academic Excellence Scholarship 2022–2024
- Pusan National University

Foreign Language Special Lecture Consignment Program Scholarship Jan 2021
- Pusan National University

Creative Human Resources Development Project Group Scholarship Jan 2018
- Pusan National University

Selected Activities & Awards

Invited to Samsung Heavy Industries Wind Propulsion Ship Forum (travel award) July 2024
- Samsung Heavy Industries

Participating in 'Conceptual Design of Efficient Devices of a 25-tonne High-Speed Ship' Jan. 2024
- Korean Society of Shipbuilding and Marine Engineering / Coast Guard

Invited to Hyundai Heavy Industries Eco-Friendly Smart Ship Forum (travel award) Apr. 2023
- Hyundai Heavy Industries

TPC (Test Preparation Course) Certification — English Consignment Program July 2021
- Pusan National University

Teaching & Tutoring

Private Tutoring (Educational Volunteering), Mathematics Instructor South Korea
- Taught high-school mathematics as an educational volunteer for two years 2022 – 2024

Private Tutoring, Physics & Chemistry Instructor — College Entrance Preparation South Korea
- Taught physics and chemistry to ten high-school seniors preparing for the Korean college entrance exam (CSAT) over six months 2023 – 2024

Languages

Korean: Native

English: Fluent